15 Prove algebraically that the difference between the squares of any two consecutive integers is always an odd number.

Let the consecutive integers be n and n+1, n integer.

$(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2$

1 mark
Expand squares

$= 2n + 1.$

1 mark
2n+1

2n is even, so2n+1 is always odd.

1 mark
Even plus1 is odd

(Total for Question 15 is 3 marks)